Yik Yu Ng

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Education

Stanford University **Starting Fall 2026**
Master of Science in Computer Science *Stanford, CA*

McGill University **Aug 2023 – May 2026**
Bachelor of Science, First Class Honours in Computer Science *Montreal, QC*

Publications

Parameterized Fair Resource Allocation under Diversity Constraints *SIGKDD 2026*
K. Huang, **Y. Y. Ng**, L. V. S. Lakshmanan, and X. Xiao

MADS: Ensemble LLM-based Multi-Agent Debate System for Political Bias Detection *Submitted to DSAA 2026*
Y. Y. Ng, B. C. M. Fung, E. Obukhova, D. Mouheb, C. Huang, and S. Huang

GlucoSense: Non-Invasive Glucose Monitoring using Mobile Devices *MobiCom 2025*
N. Sharma, M. Bebawy, **Y. Y. Ng**, and M. Hefeeda

Research Experience

Research Trainee **May 2026 – Present**
Mila – Quebec AI Institute *Montreal, QC*

- Developing a privacy-preserving multimodal large language model (MLLM) agent system for personalized multiple-sclerosis lesion-activity prognosis from clinical narratives and MRI.
- Built a local retrieval–synthesis–critique pipeline with grounded evidence (AI co-scientist).
- Current work analyzes systematic failure subgroups using multimodal decision embeddings.
- Supervised by Tal Arbel

Research Assistant **Jun 2025 – Jun 2026**
University of British Columbia *Vancouver, BC*

- Proposed a fair resource-allocation framework with a probabilistic recommendation prior, optimizing global welfare while preserving user preferences under diversity constraints.
- Developed explainable graph-learning methods for psychosis prediction from conversational text, including speech-graph construction, classification, and graph frequency interpretation.
- Supervised by Laks V. S. Lakshmanan and Xiaokui Xiao.

Student Researcher **Jan 2025 – Oct 2025**
McGill University *Montreal, QC*

- Developed MADS, an ensemble multi-agent LLM debate system for detecting political bias in news articles, with paragraph-embedding-based debate routing.
- Outperformed supervised-learning baselines and larger LLMs; contributed a dataset for political-article classification.
- Supervised by Benjamin C. M. Fung and Elena Obukhova.

Research Experience (continued)

Research Assistant, NSERC USRA **May 2024 – Dec 2024**
Simon Fraser University *Burnaby, BC*

- Designed a mobile-device NIR data-collection protocol for non-invasive glucose concentration prediction.
- Applied deep learning to hyperspectral reconstruction and used model explanations to identify informative wavelengths for the sensing protocol.
- Supervised by Mohamed Hefeeda, Network and Multimedia Systems Lab.

Honours and Awards

Synechron Scholarship in Computer Science – Synechron Canada **Dec 2025**

- $2,500 CAD scholarship awarded by McGill's Faculty of Science for outstanding academic merit in the Honours Computer Science program.

Research Funding – DMaS Lab, McGill University **Feb 2025**

- $16,000 CAD grant supporting research at McGill's Data Mining and Security Lab.

NSERC Undergraduate Student Research Award **May 2024**

- $11,500 CAD in research support for work at Simon Fraser University.

YP Heung Foundation Post-Secondary Award **Jul 2023**

- $5,000 CAD award as one of the top six Faculty of Science students at Langara College.

References

Gladys Wong, Dr. sc. Techn. ETH Zürich
Instructor, Algorithms and Data Structures II; Department of Computer Science, Langara College

Elena Obukhova, PhD
Associate Professor of Strategy and Organization, Desautels Faculty of Management, McGill University
Independent research supervisor; faculty profile

Yvonne Leung, CFA
Head of Managed Solutions Asia, J.P. Morgan Private Bank